Instagram Reel Virality Probability

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1 Problem Description

In the past decade social media has become one of the greatest drivers of culture. What goes viral on social media is highly impactful to society and what is popular. Instagram is one of the largest of the social media platforms, making it one of the most influential drivers of culture. In the past decade, vertical short-form video platforms, such as Instagram Reels, have become one of the most popular ways of consuming social media content. Being able to go viral on social media is by extension the ability to shape culture. This is why it is important to determine which characteristics of an Instagram Reel are most associated with virality, and how does observing these characteristics change the probability that a Reel is viral? This is important because it shapes generations' culture and shows what is valued in our society.

2 Data Sourcing and Processing

2.1 Data Source

We used What Makes an Instagram Reel Go Viral Dataset from Kaggle. This dataset includes 100,000 synthetic samples based on real data and 26 features including posting time, video length, virality score, and more.

2.2 Preprocessing

We found that the dataset contained no missing values. We excluded features that were not relevant to answering our question, such as reel_id and creator_id.

We chose features based on what we intuitively thought would contribute to Reel virality. We excluded likes, comments, and shares because they measure engagement outcomes that may overlap with the virality score. This allowed us to narrow down the features we needed to answer our question.

The features we selected were: creator_followers, hook_strength_score, hashtags_count, trending_audio, retention_rate, impressions, and virality_score.

We then converted the selected features into discrete binary variables to simplify the Bayesian Network's state space. For each numeric feature, we assigned a value of 1 when its value was at or above the chosen threshold and 0 otherwise. We used the 99.5th percentile of virality_score to define a Reel as viral for this study. For the other numeric features, we used their respective dataset medians (the 50th percentile). The trending_audio feature was already Boolean, so we cast it directly to an integer, mapping true to 1 and false to 0. The table below shows all of the discrete binary variables we generated:

Preprint.

Binary variable	Original feature	Threshold for a value of 1
bin_viral	virality_score	99.5th percentile
strong_hook	hook_strength_score	Median (≈ 0.499020)
many_hashtags	hashtags_count	Median (15)
many_impressions	impressions	Median (49,917,048.5)
many_followers	creator_followers	Median (2,500,456)
trending_audio	trending_audio	True
high_retention	retention_rate	Median (≈ 0.501637)

These transformations reduce each variable to two possible states, simplifying conditional probability estimation and inference in our Bayesian network.

3 Modeling and Inference

3.1 Model Structure and Assumptions

3.1.1 Model 1: Baseline

Using the 7 selected features, we first build a baseline Bayesian network. In this baseline, every predictor influences virality directly and the predictors do not influence each other. Later models relax this assumption while keeping `bin_viral` as the outcome, so it never has outgoing edges.

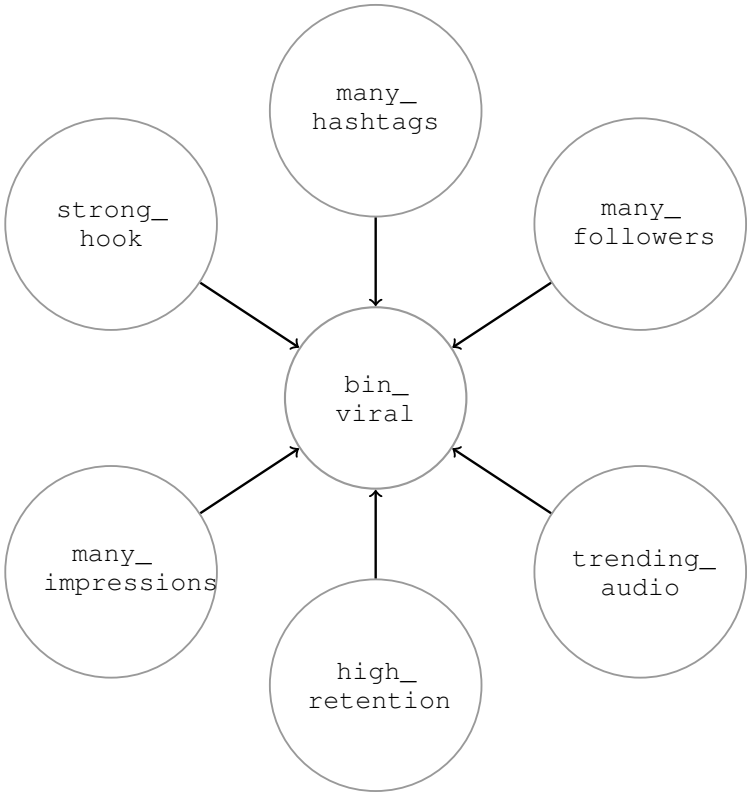


Figure 1: Baseline Bayesian network with binarized Reel features as direct predictors of the binary virality target.

The network contains six binary predictor variables: `strong_hook`, `many_hashtags`, `many_impressions`, `many_followers`, `trending_audio`, and `high_retention`. The target variable is `bin_viral`, which is 1 when `virality_score` is at or above its 99.5th percentile and 0 otherwise. The notebook binarizes each continuous predictor at its median; `trending_audio` is encoded as 1 for true and 0 for false.

As shown in Figure 1, each predictor has a directed edge to `bin_viral`, and there are no edges between predictors. Let $X_1, \ldots, X_6$ denote the predictor variables and let Y denote `bin_viral`. The joint distribution represented by this structure factorizes as

$$P(X_1, \ldots, X_6, Y) = \left(\prod_{i=1}^{6} P(X_i) \right) P(Y \mid X_1, \ldots, X_6).$$

Thus, the model estimates the probability of the binary virality outcome given the observed feature values. The structure assumes that the predictors are marginally independent; this is the baseline's main simplifying assumption, and it may not hold for correlated features. The arrows describe the network's assumed dependencies and do not imply causal relationships.

We will estimate each root variable's probability table from its observed category frequencies in the training data. The conditional probability table for `bin_viral` would have $2^6 = 64$ rows, one for each combination of predictor values. For each row, the probability of each target value is estimated from the relative frequency of that value among training examples with the corresponding predictor combination. The notebook currently performs the binarization but does not yet estimate these probability tables.

3.1.2 Model 2: Feature-Dependency Model

The second model keeps every direct edge into `bin_viral` from the baseline, but adds edges between predictors where one feature plausibly affects another. A larger follower base, trending audio, and more hashtags can each increase how many people see a Reel, so `many_followers`, `trending_audio`, and `many_hashtags` point to `many_impressions`. A strong hook keeps viewers watching, so `strong_hook` points to `high_retention`.

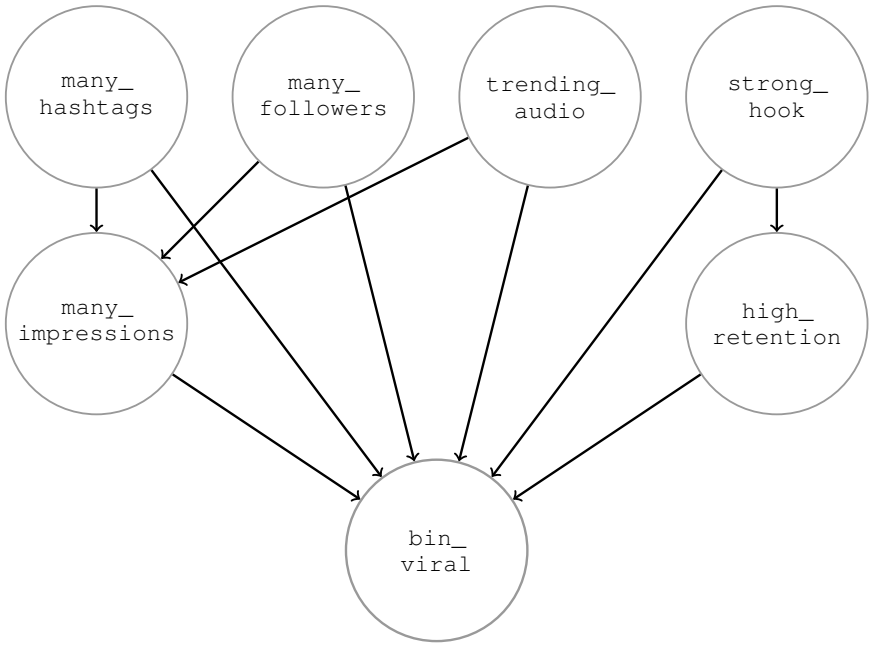


Figure 2: Model 2: the baseline with added dependencies between predictors.

For readability, let H, T, F, A, R, I, and Y denote `strong_hook`, `many_hashtags`, `many_followers`, `trending_audio`, `high_retention`, `many_impressions`, and `bin_viral`. Model 2 factorizes as

$$P(H)\,P(T)\,P(F)\,P(A)\,P(I \mid T, F, A)\,P(R \mid H)\,P(Y \mid H, T, F, A, R, I).$$

The table for `bin_viral` still has $2^6 = 64$ rows, but the predictors are no longer assumed to be independent of each other. Comparing this model with the baseline tests whether that assumption matters.

3.1.3 Model 3: Two-Stage Funnel Model

The third model treats virality as the end of a funnel. Creator and content features first determine intermediate outcomes, and only those intermediate outcomes point to `bin_viral`. As in Model 2, `many_hashtags`, `many_followers`, and `trending_audio` point to `many_impressions`, and `strong_hook` points to `high_retention`.

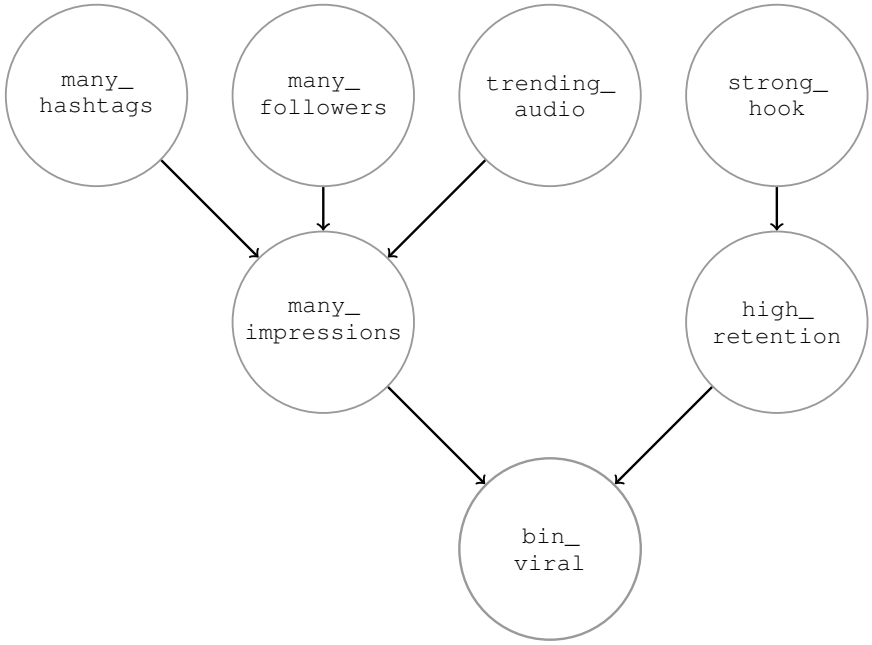


Figure 3: Model 3: creator and content features influence virality only through impressions and retention.

Model 3 factorizes as

$$P(H)\,P(T)\,P(F)\,P(A)\,P(I \mid T, F, A)\,P(R \mid H)\,P(Y \mid I, R).$$

The table for `bin_viral` shrinks from 64 rows to 4, so each row is estimated from far more training examples. The model also assumes that once impressions and retention are known, the remaining features give no further information about virality.

3.1.4 Model 4: Creator-Controlled Model

The fourth model keeps only the features a creator chooses when posting a Reel: `strong_hook`, `many_hashtags`, and `trending_audio`. Follower count, impressions, and retention are excluded because they depend on the audience or on how the Reel performs after posting. This model answers a practical question: which choices a creator makes are associated with a higher chance of going viral.

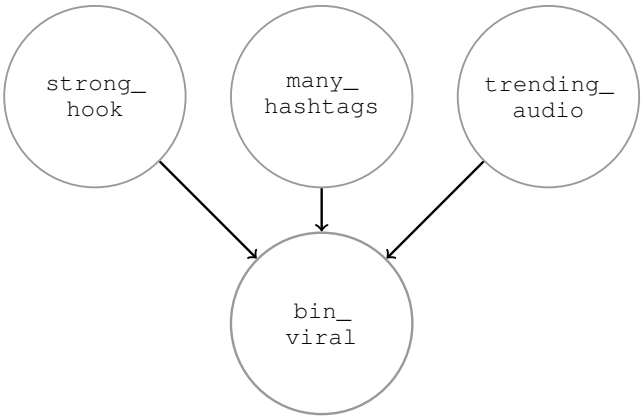


Figure 4: Model 4: only creator-controlled features predict virality.

Model 4 factorizes as

$$P(H)\,P(T)\,P(A)\,P(Y \mid H, T, A),$$

and the table for `bin_viral` has $2^3 = 8$ rows.

3.1.5 Model 5: Latent Appeal Model

The fifth model adds a hidden binary variable `appeal`, denoted Z, that represents how engaging a Reel is overall. Appeal is never observed in the data. The creator-controlled features point to `appeal`, and `appeal` points to both `high_retention` and `bin_viral`. Retention is included as a second child of `appeal` because it is observed and gives the model more information for estimating the hidden variable.

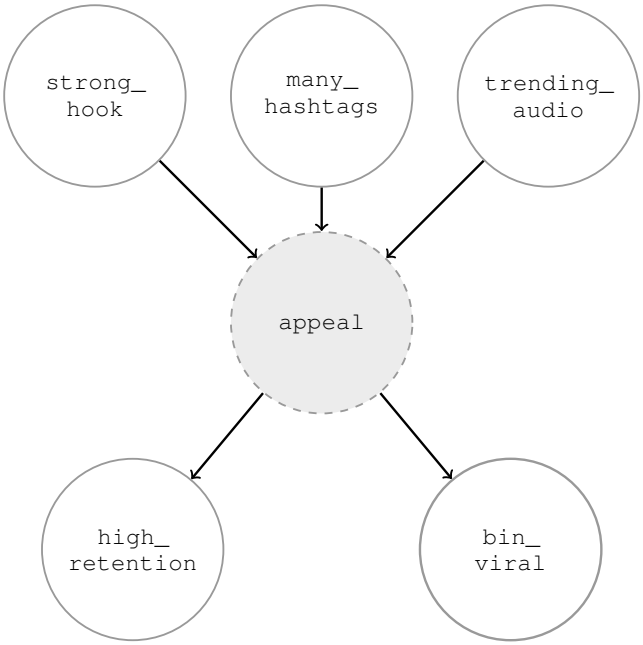


Figure 5: Model 5: a hidden appeal variable (dashed) links creator choices to retention and virality.

Model 5 factorizes as

$$P(H)\,P(T)\,P(A)\,P(Z \mid H, T, A)\,P(R \mid Z)\,P(Y \mid Z).$$

Because Z is never observed, its probability tables cannot be estimated by counting. Instead, they are learned with the Expectation-Maximization (EM) algorithm, which alternates between inferring Z for each Reel and re-estimating the tables until the log-likelihood converges.

3.2 Inference Algorithm

We intend to use approximate inference to estimate the probability that a Reel is viral given observed characteristics.

[TODO: Specify the implemented inference algorithm, evidence and query variables, algorithm settings, and implementation details.]

4 Results and Discussion

4.1 Experimental Configurations

[TODO: Compare multiple configurations, such as inference settings, feature subsets, discretization thresholds, or network structures. Describe the evaluation procedure and explain observed performance differences.]

4.2 Quantitative Results

[TODO: Report measured results with suitable metrics, tables, or figures. Explain the metrics and compare performance across configurations.]

4.3 Qualitative Insights

[TODO: Explain what the model captures about the data, how observing features changes virality probabilities, and whether learned conditional probability tables align with intuition.]

4.4 Convergence and Scalability

[TODO: Discuss training or inference stability, convergence, runtime, and computational cost as the dataset or network grows.]

5 Conclusion

[TODO: Summarize key findings and model performance. Discuss limitations, including the use of synthetic data, and propose extensions or improvements.]

6 Reflections & Contributions

6.1 Advice for Future Students

[TODO: Share practical suggestions and lessons learned from the project.]

6.2 Individual Contributions and Reflections

Iban Rosales *[TODO: Write 1–2 sentences describing your contributions and a brief individual reflection on the project.]*

Ethan Fang *[Writing latex for the project report.]*

Conner Satawidinata *[TODO: Write 1–2 sentences describing your contributions and a brief individual reflection on the project.]*

Benjamin Miller *Wrote a lot of the project writeup and assisted with overall project ideas.*

6.3 Use of Generative AI

Gen AI was used to assist in the generation of Python and Latex code in order to translate our ideas into code. All ideas where discuessed and choosen by humans, and we used AI to assit us in implementation detials. We manaully reviewed all code and outputs to verify it matches our intent.

7 References

[TODO: Cite all external papers, datasets, and tools used in the project. Replace this placeholder with the actual reference entries.]